Overview of Kalman Filter Theory and Applications

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Segments 9-12: Kalman Filter Applications: Terrestial Navigation

- Some Practical Considerations (Numerical)
- The Navigation Problem and Reference Frames
- Simple Examples of Kalman Filtering Applied to Navigation
- Inertial Error Dynamics
- The Basic 9-State Kalman Filter

- Challenges of INS Filters

Segment 9
Some Practical Considerations (Numerical)
Topics

- Kalman Filter Theory vs. Reality
- Numerical Stability
- Round-Off Errors
- Matrix Inversion
- UD Factorization

Kalman Filter Theory vs. Reality

- Kalman theory based on multiple assumptions
  - All states are modeled correctly
  - Noise parameters are accurately known
  - Initial values are appropriate
  - Process and measurement noise are both Gaussian
  - Process noise and measurement noise are uncorrelated
  - Measurements are taken at regularly scheduled intervals

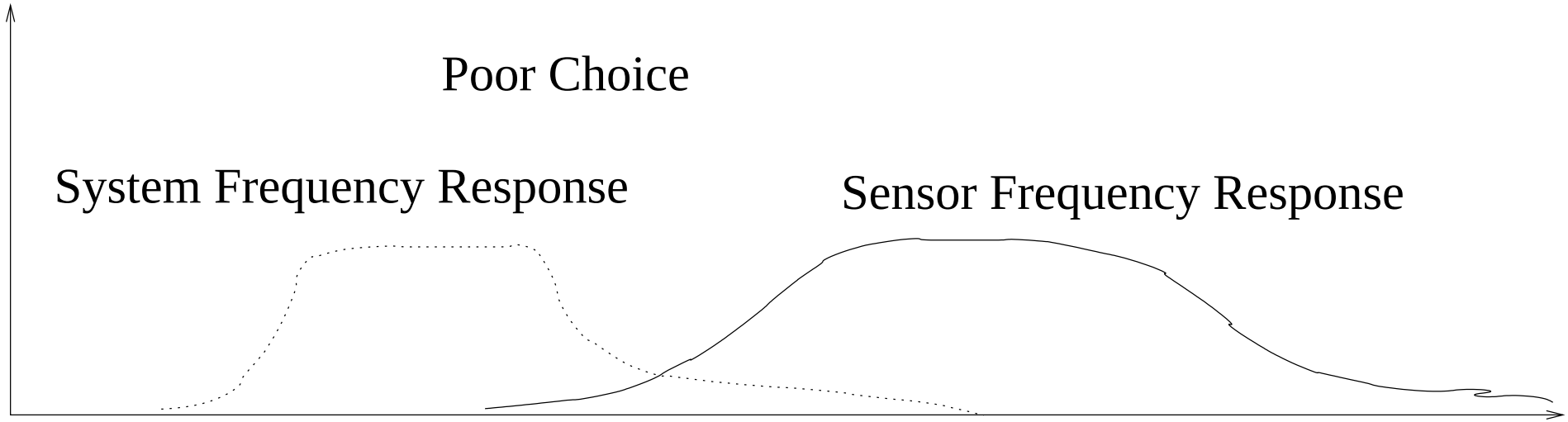
Kalman Filter Theory vs. Reality

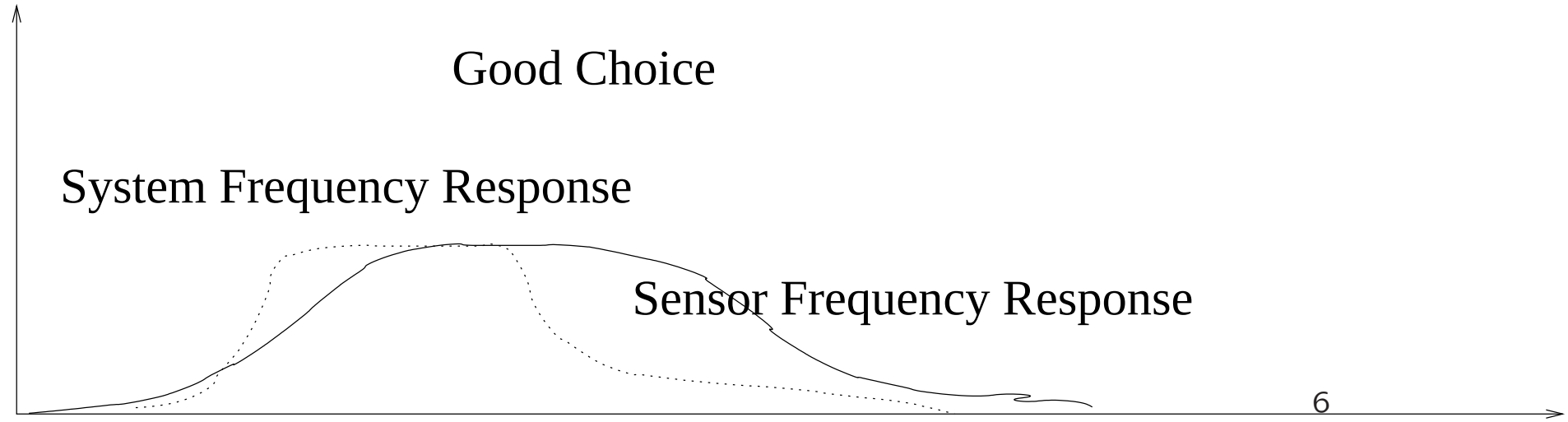
- How do you select appropriate sensors for the problem at hand?

- Some obvious general rules:
  - the lower the sensor noise characteristics, the more accurate the estimates (narrow sensor noise variance)
  - the more the merrier, i.e., the more sensors the better
  - maximize the number of directly observable states

Kalman Filter Theory vs. Reality

- Some less obvious general rules:
  - short-term / long-term complementarity (e.g, GPS vs. INS)
  - frequency response of sensor should match that of system
  - noise spectrum of sensor should be maximally disjoint from that of system


Poor Choice
System Frequency Response
Sensor Frequency Response


Good Choice
System Frequency Response
Sensor Frequency Response

Kalman Filter Theory vs. Reality

- Choose sensors to match states.
  E.g., consider the state model

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

  with the measurement model

$$z = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + v$$

  Here x_1 is observable but x_2 is not.

Kalman Filter Theory vs. Reality: Sensor Selection

- Examine the H matrix that you would use with this sensor!

- If it renders some states unobservable, then consider adding additional sensors for those states, or replacing this sensor with one that is more favorable.

Filter Divergence

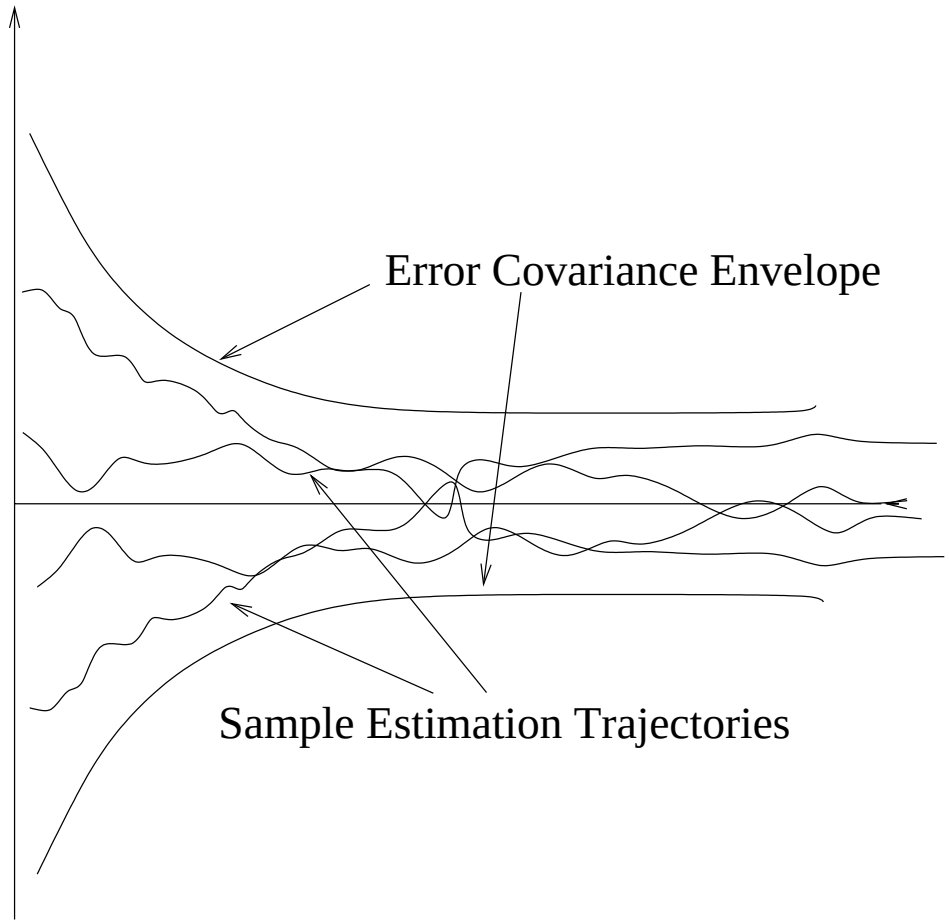
- Kinds of Divergence
- Causes and Remedies

Kinds of Divergence

- Mean estimation error is supposed to converge to 0
  - $\tilde{x} = \hat{x}^{+}(k) - x(k)$
  - $\mathsf{E}\left[\tilde{x}\right] \rightarrow 0$
  - Estimate is unbiased: $\mathsf{E}\left[\hat{x}_k\right] = x_k$
- Covariance matrix for estimation error is same as state estimation error covariance matrix: $P_k^{+} = \mathsf{E}\left[\tilde{x}^{+}(k)[\tilde{x}^{+}(k)]^{\top}\right]$

Kinds of Divergence

- For optimal filter, P will indicate convergence or divergence.
  - One common form of divergence due to round-off will send P to zero
- For suboptimal filters, P will not necessarily indicate divergence
  - P may seem to indicate nominal steady-state values and yet the state estimate may go diverge from the true value


Error Covariance Envelope
Sample Estimation Trajectories

Kinds of Divergence

- Going to sleep: error covariance becomes so small that measurements are no longer believed by the filter
- Loss of symmetry in P
- Loss of positive definiteness: negative numbers on the diagonal!

Numerical Stability

- System stability: Bounded input results in bounded output
- Numerical stability: small errors at each computation step do not grow without bound
- Matrix Conditioning

Matrix Conditioning

- Ill-Conditioned vs. Well-Conditioned

- Condition number of matrix:

$$\text{cond}(M) = \frac{\lambda_{\text{max}}}{\lambda_{\text{min}}}$$

  where λ_{max} is maximum eigenvalue of M, while λ_{max} is the minimum eigenvalue of M.

Matrix Conditioning

- If cond(M) ≈ 1, then M is well-conditioned for matrix inversion

- If cond(M) $>> 1$, then M is ill-conditioned for inversion

Round-Off Errors

- Computer round-off in floating point arithmetic
- Unit round-off value is largest number ε such that $1 + \varepsilon = 1$
- In Matlab, the global symbol "eps" stores the unit round-off value

Round-Off Errors

- Round-off errors generally plague the classical covariance update algorithm

$$P_k^+ = [I - K_k H_k] \, P_k^-$$

- Alternative, called the Joseph form, is more robust:

$$P_k^+ = [I - K_k H_k] \, P_k^- \, [I - K_k H_k]^\top + K_k R_k K_k^\top$$

This form in turn leads to the De Vries mechanization of the Kalman gain equation. To see this, first expand the Joseph form (omitting subscripts for clarity):

$$P(+) = P(-) - KHP(-) - P(-)H^\top K^\top + KHP(-)H^\top K^\top + KRK^\top.$$

Define

(1) $$F = PH^\top$$

and

(2) $$D = HF + R.$$

We note that D is symmetric, because

$$\begin{aligned}
D^\top &= [HF + R]^\top \\
&= [HPH^\top + R]^\top \\
&= [HPH^\top]^\top + R^\top \\
&= [H^\top]^\top P^\top H^\top + R^\top \\
&= HPH^\top + R \qquad \text{(because } P \text{ and } R \text{ are symmetric)} \\
&= HF + R \\
&= D.
\end{aligned}$$

The gain in standard Kalman gain equation

$$K_k = P_k^- H_k^\top \left[H_k P_k^- H_k^\top + R_k \right]^{-1}$$

can be written as

$$K = FD^{-1}.$$

Using Eqs. (1) and (2), we compute

$$\begin{aligned} P(+) &= P(-) - KF^\top - FK^\top + KDK^\top \\ &= P(-) - K\left[F^\top - \frac{DK^\top}{2}\right] - \left[F - \frac{KD}{2}\right] K^\top. \end{aligned}$$

Letting $G = F - \dfrac{KD}{2}$, we obtain

$$P(+) = P(-) - GK^\top - \left[GK^\top\right]^\top. \tag{3}$$

The De Vries mechanization implements Eq. (3) by subtracting the ij^{th} element of $GK^\top$ from both the ij^{th} and ji^{th} elements of $P(-)$. The De Vries form of the covariance update equation is valid for any arbitrary gain matrix K, not just the optimal gains computed from Eqs. (1) and (2).

Matrix Inversion

- Matrix inversion is always risky due to potential ill-conditioning
- One way to avoid this is to process measurements sequentially as scalars
- Or at least as low-dimensional matrices

UD Factorization

- Based on Cholesky Decomposition

- Unit Triangular Matrices: U has 1's on the diagonal and all 0's below the diagonal. E.g.,

$$U = \begin{bmatrix} 1 & 5 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

UD Factorization

- Factor $HPH^\top + R$ as $UDU^\top$, where U is unit upper triangular and D is diagonal.

- This then allows us to "invert" $HPH^\top + R$ by solving the equation unit upper triangular equation $UDU^\top X = H$ for the matrix X, which will yield $X = \left[HPH^\top + R\right]^{-1} H$

- This last expression is used on the covariance update equation in KH

Exercises

1. Let $v = [1 - 2]^\top$. Compute the positive definite, symmetrix matrix $M = vv^\top$

2. Use the following algorithm (Grewal & Andrews) to factor M into $UDU^\top$ form:

```
for j=m:-1:1
    for i=j:-1:1
        covar=P(i,j);
        for k=j+1:m
            covar=covar-U(i,k)*D(k,k)*U(j,k);
        end
        if i==j
            D(j,j)=covar;
            U(j,j)=1;
        else
            U(i,j)=covar/D(j,j);
        end
    end
end
```

Segment 10
The Navigation Problem and Reference Frames
Topics

- Basic Navigation Problem
- Earth Referenced Navigation
- Coordinate Frame Conversion
- Basic Strapdown Navigation Equations

Basic Navigation Problem

- Where am I?
- Where am I headed?
- How fast am I going?

Basic Navigation Problem

- What's my position?
- What's my heading and attitude?
- What's my velocity?

Solution in Inertial Space

- In inertial space, it's no problem!

- Just integrate the acceleration once to get velocity, twice to get position.

$$v(T) = v(0) + \int_0^T \frac{d^2 r}{dt^2} dt$$

$$r(T) = r(0) + \int_0^T v(T) dt$$

- And integrate the angular rate once to get attitude.

Solution in Inertial Space

- Solution easy in inertial space because ...
  - inertial instruments sense acceleration and rotational rate with respect to inertial coordinate frame
  - inertial frame attached to sensor platform
  - Note: You still need to rotate accelerations from body to inertial coordinates prior to integration (strapdown)
- But we don't necessarily want the answer to our questions expressed in terms of inertial coordinates!

Earth Referenced Navigation

- Terrestial navigation seeks to answer nav questions with respect to earth
  - Global earth reference
  - Local or relative navigation
- Complications:
  - Earth surface is curved (ellipsoid)
  - Earth is turning (can usually ignore motion about the Sun for terrestrial nav)

Coordinate Frames

- Inertial
- Earth Fixed
- Local Level
- Body

Coordinate Frames

- Technically speaking, all frames are earth-centered
- Practically speaking, we think of the Body frame as centered in the body of the vehicle (usually at center of gravity).
- Local Level is thought to be located at surface of earth directly beneath the vehicle.

Inertial Reference Frames

- Non-Rotating
- Non-Accelerating
- Fixed or moving uniformly with respect to another inertial frame (fixed stars)

Inertial Reference Frames

- Nav problem "easy" in inertial frames
- Integrate gyro rates once to get attitude (rotational motion is absolute)
- Rotate accelerometer outputs to inertial frame using attitude matrix, then integrate to get inertial velocity; integrate velocity to get position.
- Would still have to contend with instrument imperfections.

Inertial Reference Frame Integrations

(4) $$\ddot{r}(t) = a_{\mathsf{sf}} + \text{other external forces}$$

(5) $$v(T) = \dot{r}(T) = v(0) + \int_0^T a_{\mathsf{sf}} dt + \int_0^T (\text{other external forces}) dt$$

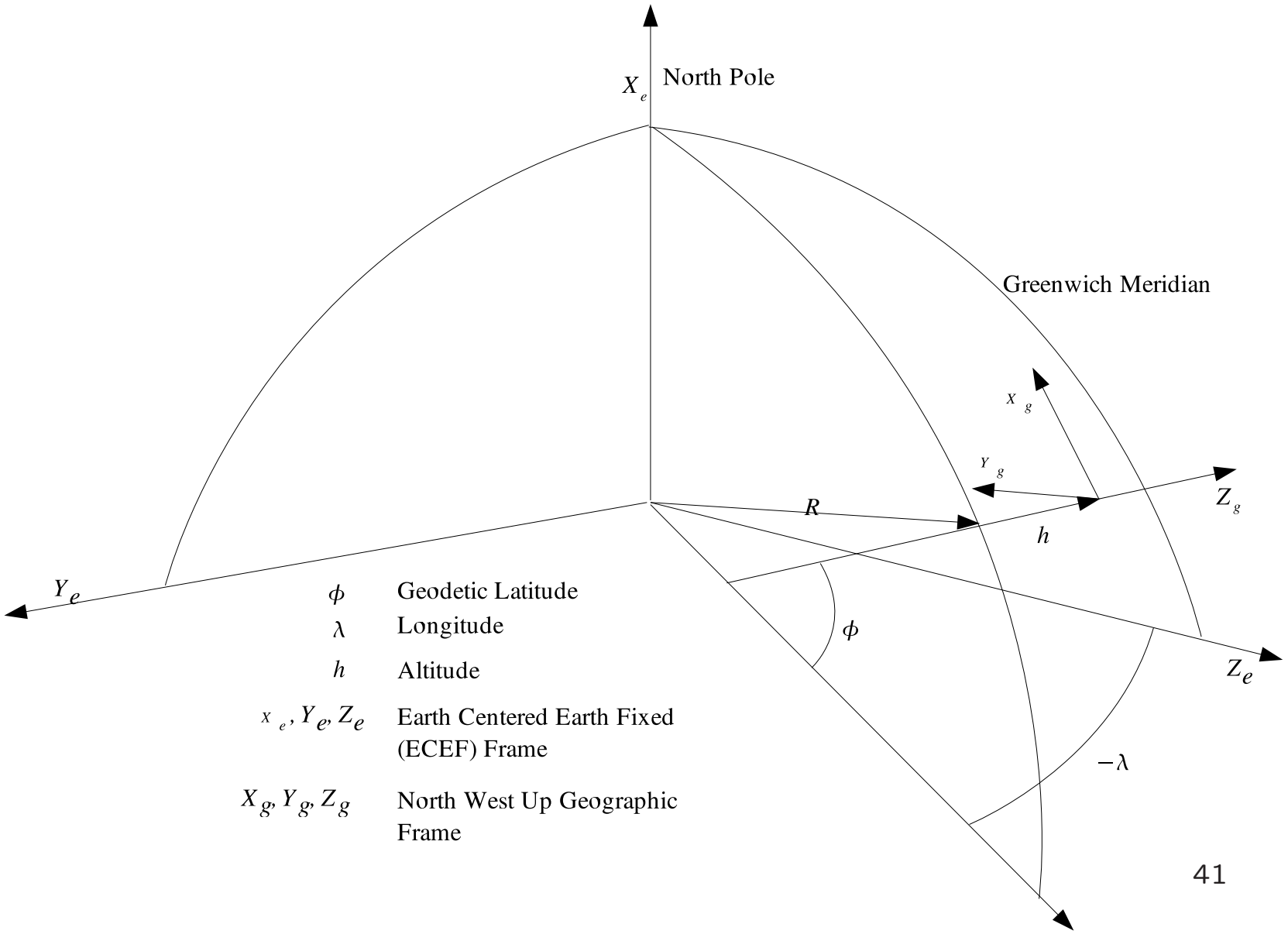
(6) $$r(T) = r(0) + \int_0^T v dt$$

Earth Fixed = ECEF = Earth Centered Earth Fixed

- Rotating frame, i.e., non-inertial
- Cartesian (X, Y, Z in meters)
- Spherical (latitude ϕ, longitude λ, altitude h)
  - geodetic

Local Level

- Tangent to earth's surface but translated to center of earth.
- Vertical Z-axis is normal to earth's surface
- X, Y-axes are level
- Several kinds: Y north-slaved, wander (0 transport rate) or free azimuth
- Y makes wander angle α with respect to true north (ENU=East, North, Up).



ϕ	Geodetic Latitude
λ	Longitude
h	Altitude
X_e, Y_e, Z_e	Earth Centered Earth Fixed (ECEF) Frame
X_g, Y_g, Z_g	North West Up Geographic Frame

Body Frame

- Attached to vehicle

- For aircraft:
  - X out the nose
  - Y out the right wing
  - Z down

Coordinate Frame Conversion

- All frames are earth-centered, hence differ only by rotation
- Rotation represented by direction cosine matrices (DCM)
  - Represents the angles made by each of the basis vectors of one frame with those of the other
- C_A^B = DCM transformation from frame A to frame B

Coordinate Frame Conversion

- "Direction Cosine" comes from dot product:

$$\langle x, y \rangle = x \cdot y = \|x\| \, \|y\| \cos \theta_{xy}$$

  where θ_{xy} is the angle between the two vectors x and y.

- If x and y are unit vectors, then $\|x\| = \|y\| = 1$ and we have:

$$\langle x, y \rangle = x \cdot y = \cos \theta_{xy}$$

Coordinate Frame Conversion

- If $U = \{u_1, u_2, u_3\}$ and $V = \{v_1, v_2, v_3\}$ are two sets of orthonormal basis vectors, then the DCM relating one frame to the other is given by:

$$C_U^V = \begin{bmatrix} \langle u_1, v_1 \rangle & \langle u_1, v_2 \rangle & \langle u_1, v_3 \rangle \\ \langle u_2, v_1 \rangle & \langle u_2, v_1 \rangle & \langle u_2, v_3 \rangle \\ \langle u_3, v_1 \rangle & \langle u_3, v_2 \rangle & \langle u_3, v_3 \rangle \end{bmatrix}$$
$$= \begin{bmatrix} \cos\theta_{11} & \cos\theta_{12} & \cos\theta_{13} \\ \cos\theta_{21} & \cos\theta_{22} & \cos\theta_{23} \\ \cos\theta_{31} & \cos\theta_{32} & \cos\theta_{33} \end{bmatrix}$$

Coordinate Frame Conversion

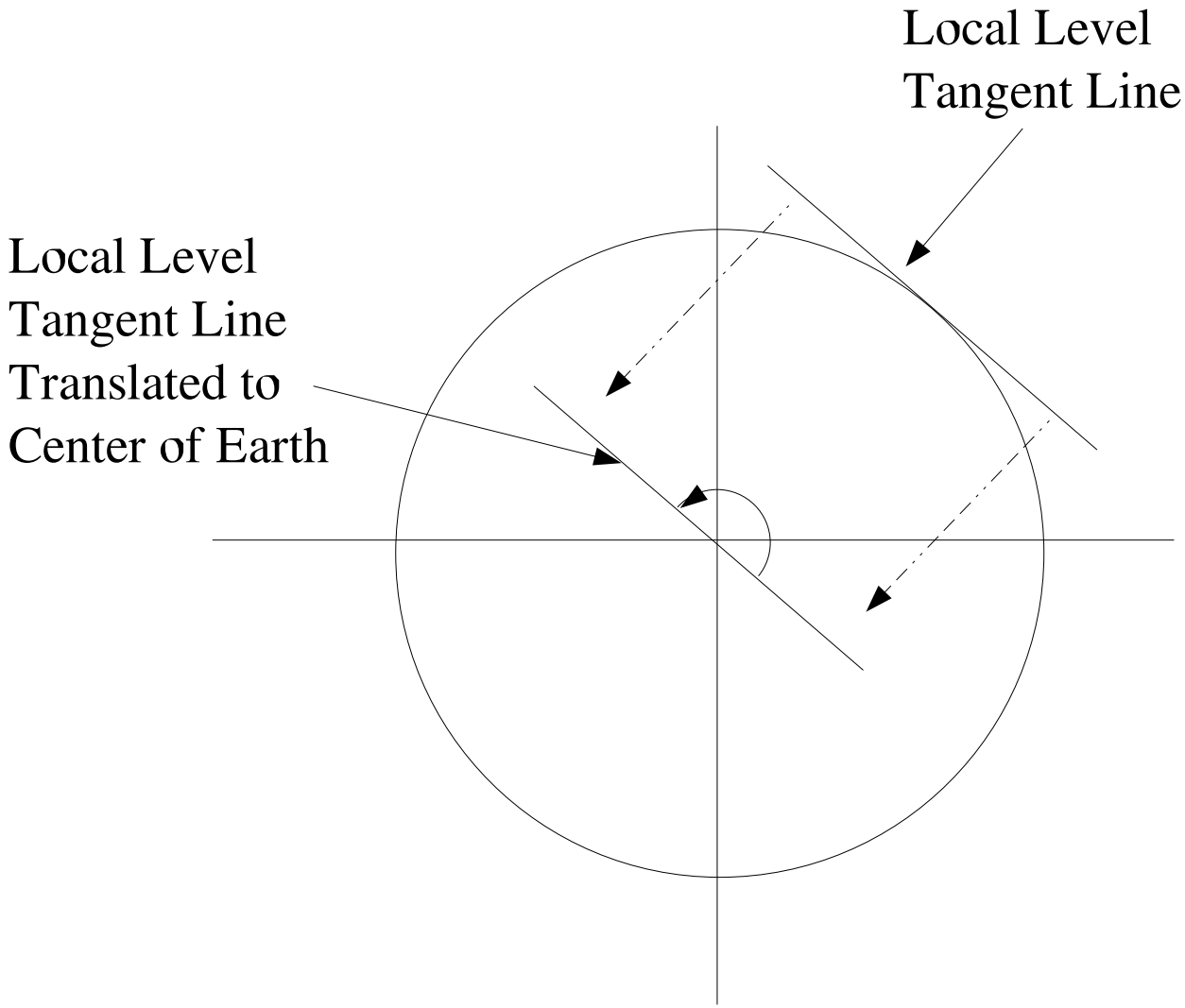
- DCMs are orthogonal matrices $\Rightarrow \left[C_A^B\right]^{-1} = \left[C_A^B\right]^\top$
- Third row (or column) is vector cross product of other two $\Rightarrow$ only six independent parameters
- DCMs satisfy: $C_B^A C_C^B = C_C^A$

Coordinate Frame Conversion: Most important DCMs

- Body to Local Level: C_B^L
  - sometimes called Body to Nav and notated C_B^N
  - yields attitude info (Euler angles for roll, pitch, yaw)
- Local Level to Earth: C_L^E
  - yields position info and also used to transform earth rate to local level

Coordinate Frame Conversion: Most important DCMs

- Visualize C_L^E's determination of position as follows:
  - As vehicle moves, local level frame changes its orienation with respect to earth's center in order to maintain local level with respect to earth's surface
  - If you think of local level frame as attached to earth's surface, you get distracted by the displacement of the frame
  - If you move local level frame to center of earth, the displacement disappears and all that remains are the rotations


Local Level
Tangent Line
Local Level
Tangent Line
Translated to
Center of Earth

Strapdown Navigation: Strapdown vs. Gimbaled

- Gimbaled INS:
  - Gyro stabilized platform
  - Gimbal assembly and torque motors
  - Resolvers, slip rings, bearings, etc. to control platform
  - Goal: Maintain stable, locally level or inertial reference frame

Strapdown Navigation: Strapdown vs. Gimbaled

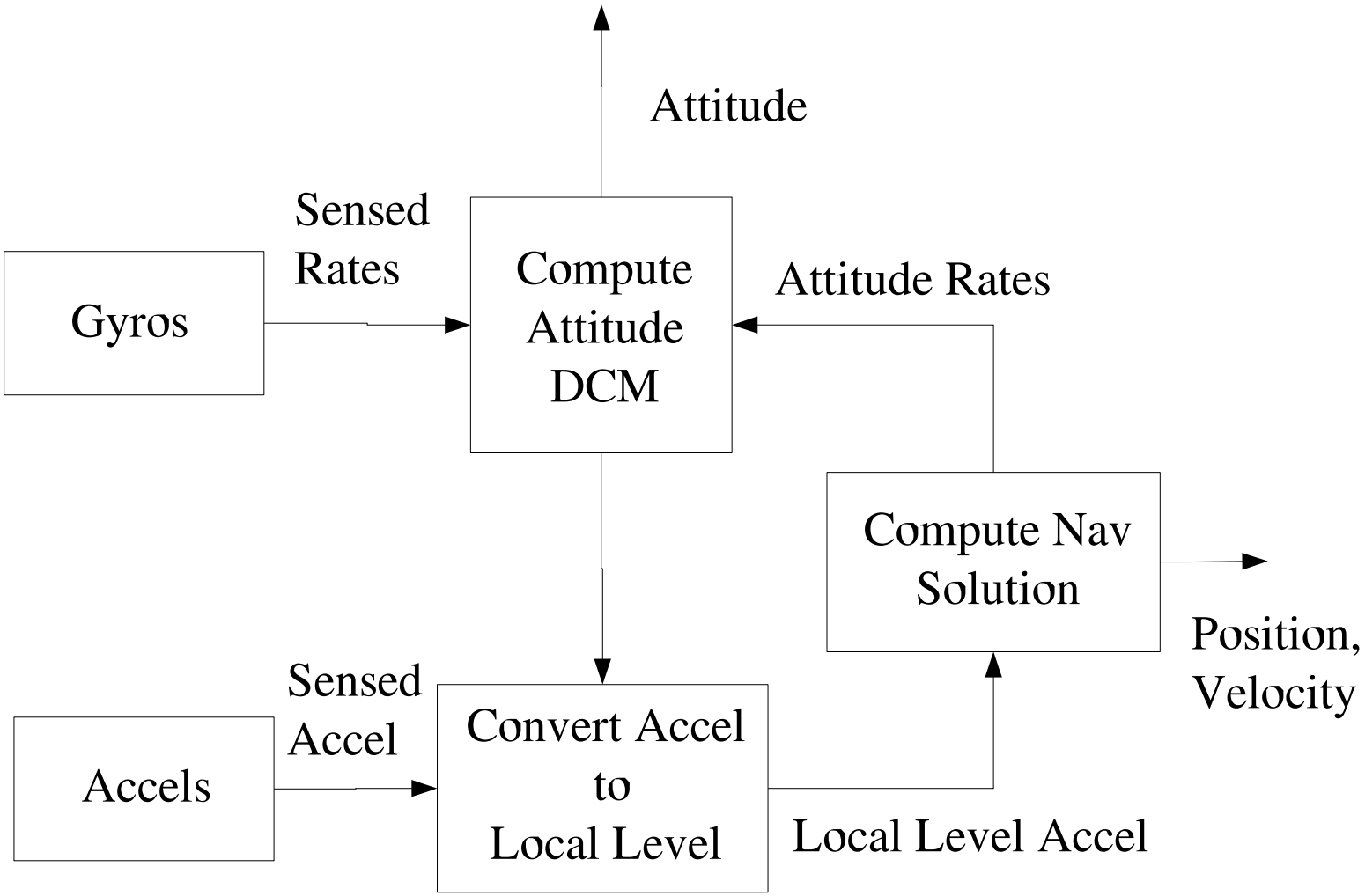
- Gimbaled INS Issues
  - Reliability and Cost: Lots of moving parts!
  - Body rate and acceleration information not directly available
  - Rate signals are generally noisy and have low bandwidth

Strapdown Navigation: Strapdown vs. Gimbaled

- Strapdown INS:
  - Inertial instruments strapped to body frame
  - Requires local level accelerations to be computed using direction cosine matrix C_B^L
  - C_B^L derived from gyro-sensed body rates
  - Strapdown systems can be implemented with either rate gyros or attitude gyros

Strapdown Navigation: Strapdown vs. Gimbaled

- Strapdown INS Advantages
  - Immediate detection of body rates and accelerations
  - Fewer moving parts
  - Potential redundancy advantages
- Gimbaled systems are nearly obsolete



Strapdown INS Block Diagram

Basic Strapdown Navigation Equations

- Need to compute attitude, velocity, and position
- I.e., need to compute C_B^L and C_L^E
- These DCMs are dynamic quantities
- C_B^L determines attitude and can be rapidly varying
- C_L^E determines position and is slowly varying (relative to attitude dynamics)

Rotating Coordinate Frames

- We first need to understand how an arbitrary vector and its time derivatives in one coordinate frame are expressed in another frame that is rotating with respect to the first.

- Let A and B denote two coordinate frames and assume that B is rotating with respect to A.

- We use the notation ω_{AB}^{A} to express the direction and magnitude of the rotation vector as seen in the A frame.

Rotating Coordinate Frames

- Every rotation vector can also be expressed as a skew symmetric matrix.

- If ω is a rotation vector (without mentioning the frames involved), then

$$\Omega = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}.$$

Rotating Coordinate Frames

- Since multiplication of a vector by the skew symmetric matrix equivalent is the same thing as left multiplying the cross product of the original vector, we can write:

$$\Omega v = \omega \times v$$

- Therefore the notation $[\,\omega\times\,] = \Omega$ is common*

*One must be careful in making this identification since left matrix multiplication is associative while left cross product multiplication is not.

Rotating Coordinate Frames

- Let v be an arbitrary vector in 3-space.
- Let $\left\{e_A^1, e_A^2, e_A^3\right\}$ be orthonormal basis vectors in the A frame
- Let $\left\{e_B^1, e_B^2, e_B^3\right\}$ be orthonormal basis vectors in the B frame*.

*We reverse the roles of superscripts and subscripts for basis vectors so that we may employ the Einstein summation convention on occasion.

Rotating Coordinate Frames

- Thus, we can write v as $\sum_{i=1}^{3} v_i^A e_A^i$ and also $\sum_{i=1}^{3} v_i^B e_B^i$.

- This is one and the same vector expressed in two different coordinate frames*.

*A vector is a linear combination of basis vectors and not necessarily an ordered triple of real numbers.

Rotating Coordinate Frames

- Note that, with respect to an arbitrary inertial reference frame, we have for the time derivative:

$$
\begin{aligned}
v &= \sum_{i=1}^{3} v_i^A e_A^i = \sum_{i=1}^{3} v_i^B e_B^i \\
\Rightarrow \dot{v} &= \frac{d}{dt}\left[\sum_{i=1}^{3} v_i^A e_A^i\right] = \frac{d}{dt}\left[\sum_{i=1}^{3} v_i^B e_B^i\right] \\
\Rightarrow & \sum_{i=1}^{3}\left(\dot{v}_i^A e_A^i + v_i^A \dot{e}_A^i\right) = \sum_{i=1}^{3}\left(\dot{v}_i^B e_B^i + v_i^B \dot{e}_B^i\right)
\end{aligned}
$$

Rotating Coordinate Frames

- In the special case that their respective coordinate frames are not rotating relative to one another, each $\dot{e}^i_A = 0 = \dot{e}^i_B$.

- Nevertheless, regardless of rotation, we must always have:

(7) $$\sum_{i=1}^{3} \dot{v}_i^A e_A^i = \sum_{i=1}^{3} \dot{v}_i^B e_B^i.$$

(Note that, in general, $\dot{v}_i^A \neq \dot{v}_i^B$.)

Rotating Coordinate Frames

- Let frame B rotate with respect to frame A at a constant rotational rate ω_{AB}^{A} as expressed in A frame coordinates.

- Since B is not rotating with respect to itself, let us agree to label the quantity in Eq. (7) as $\dot{v}^{B}$.

- We'll assume that A is an inertial frame and thus $\dot{v} = \dot{v}^{A}$.

Rotating Coordinate Frames

- When viewed from the A frame, however, it is not true that each of the $\dot{e_i}^B$ is zero, since these unit vectors are slaved to the B frame and B is rotating with respect to A at rate ω^A_{AB}.

- Therefore, we have:

$$\begin{aligned} \dot{v}^A &= \sum_{i=1}^{3} \dot{v}_i^A e_A^i + \sum_{i=1}^{3} v_i^B \dot{e}_B^i \\ &= \sum_{i=1}^{3} \dot{v}_i^B e_B^i + \sum_{i=1}^{3} v_i^B \dot{e}_B^i \quad \text{(by Eq. (7))} \\ &= \dot{v}^B + \sum_{i=1}^{3} v_i^B \dot{e}_B^i. \end{aligned} \tag{8}$$

- The last sum on the right side of Eq. (8) can be rewritten as the cross product $\omega^A_{AB} \times v$, because $\dot{e}^i_B = \omega^A_{AB} \times e^i_B$, and

$$\begin{aligned}\sum_{i=1}^{3} v_i^B \dot{e}_B^i &= \sum_{i=1}^{3} v_i^B \omega_{AB}^A \times e_B^i \\ &= \omega_{AB}^A \times \sum_{i=1}^{3} v_i^B e_B^i \\ &= \omega_{AB}^A \times v.\end{aligned}$$

Rotating Coordinate Frames

- We thus arrive at the all-important Coriolis Equation which is used often in navigation analysis:

$$\dot{v}^A = \dot{v}^B + \omega^A_{AB} \times v^B. \tag{9}$$

Alternatively, we can express Eq. (9) in skew-symmetric matrix form using $\Omega_{AB} = [\,\omega_{AB}\times\,]$:

$$\dot{v}^A = \dot{v}^B + \Omega^A_{AB} v^B. \tag{10}$$

Note that we must use superscripts on both Ω_{AB} and v as soon as we make a commitment to a particular coordinate frame representation.*.

*Since matrices are defined in terms of their action on basis vectors, it is appropriate to specify the coordinate system. On the other hand, if Ω_{AB} is taken to be a linear transformation, then superscripts are not necessary.

Strapdown Navigation Equations

- Three differential equations
  - Equation for body to local level transformation C_B^L
  - Equation for velocity (uses C_B^L)
  - Equation for local level to earth for position DCM C_L^E

Strapdown Navigation Equations

- Two Algebraic Equations
  - Equation for converting velocity to transport rate (to be explained)
  - Equation for converting earth rate to local level coordinates

Strapdown Navigation Equations

The Three Differential Equations

(11) $$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L$$

(12) $$\dot{v}^L = C_B^L a_{\mathsf{sf}}^B + g' - \left(2\Omega_{IE}^L + \Omega_{EL}^L\right) v^L$$

(13) $$\dot{C}_L^E = C_L^E \Omega_{EL}^L$$

Strapdown Navigation Equations

The Two Algebraic Equations

(14) $$\omega_{EL}^{L} = \frac{1}{r} u_r^L \times v^L \text{ (wander azimuth } \rho_z = 0)$$

(15) $$\omega_{IE}^{L} = \left[C_L^E\right]^\top \omega_{IE}^{E}$$

Strapdown Navigation Equations

The C_B^L Equation (11)

$$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L$$

- All the Ω matrices represent rotation vector cross products in skew-symmetric matrix format
  - Ω_{IB}^B is rotational rate of aircraft in body coordinates
  - Ω_{IE}^L is earth rate in local level coordinates
  - Ω_{EL}^L is transport rate in local level coordinates

Strapdown Navigation Equations: Detour into Cross Product Matrices

- The cross product of $\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$ with any other vector v is the same as multiplying v on the left by $\Omega = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$

- $\omega \times v = \Omega v$

Strapdown Navigation Equations

The C_B^L Equation (11)

$$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L$$

- Transport rate is the rotational rate of the aircraft due to motion over the surface of the earth. Is a function of velocity.

- Ω_{IB}^B is total inertial angular rate sensed by gyros and includes effects of transport and earth rates.

- Hence transport and earth rate effects have to be subtracted off.

Strapdown Navigation Equations

The Strapdown Velocity Equation (12)

$$\dot{v}^L = C_B^L a_{\mathsf{sf}}^B + g' - \left(2\Omega_{IE}^L + \Omega_{EL}^L\right) v^L$$

- Sometimes referred to as the velocity rate equation

- a_{sf}^B is specific force sensed by accelerometers in body coordinates (because the accels are strapped to the body – hence the term strapdown)

Strapdown Navigation Equations

The Transport Rate Equation (13)

$$\dot{C}_L^E = C_L^E \Omega_{EL}^L$$

- Connected to velocity rate equation by the first algebraic equation (14):

$$\omega_{EL}^L = \frac{1}{r} u_r^L \times v^L$$

- Eq. (13) Sometimes referred to as position rate equation, because C_L^E directly implies the vehicle position

Strapdown Navigation Equations
Position Rate / Transport Rate Equations (Geographic)

- North transport rate:

(16) $$\rho_N = \frac{v_E}{R_e}\left[1 - \frac{h}{R_e} - e\sin^2\phi\right]$$

- East transport rate:

(17) $$\rho_E = -\frac{v_N}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3\cos^2\phi\right)\right]$$

where e is ellipticity (approx. $\frac{1}{298}$) and R_e is mean equatorial earth radius

Strapdown Navigation Equations
Position Rate / Transport Rate Equations (Geographic)

- Latitude Rate:

(18) $$\dot{\phi} = -\rho_E$$

- Longitude Rate:

(19) $$\dot{\lambda} = \rho_N \sec \phi$$

Strapdown Navigation Equations

- Geographic (north-slaved) mechanization has singularities at poles

- Two other approaches to local level coordinates:
  - Free azimuth: vertical component of transport rate ρ_z= additive inverse of vertical component of earth rate
  - Wander Azimuth: vertical component of transport rate ρ_z = 0

Strapdown Navigation Equations
Position Rate / Transport Rate Equations (Wander and Free Azimuth)

- X Transport Rate:

  (20)
  $$\rho_x = -\frac{v_y}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3d_{22}^2 - d_{21}^2\right)\right] - \frac{v_x}{R_e}\left(2d_{21}d_{22}\right)e$$

- Y Transport Rate:

  (21) $$\rho_y = \frac{v_x}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3d_{21}^2 - d_{22}^2\right)\right] + \frac{v_y}{R_e}\left(2d_{21}d_{22}\right)e$$

Strapdown Navigation Equations

Position Rate / Transport Rate Equations (Wander and Free Azimuth)

- Azimuth Rate:

$$\rho_z = \begin{cases} -\Omega_e d_{23}, & \text{if free azimuth;} \\ 0, & \text{if wander azimuth} \end{cases}$$

- In equations above, the matrix $\left[d_{ij}\right]$ is the direction cosine matrix C_L^E from local level (L) frame to earth centered earth fixed (E) frame.

Exercises

1. Compute direction cosine matrix C_U^V, where

$$U = \{(1,0,0),(0,1,0),(0,0,1)\}$$

and

$$V = \{(-1,0,0),(0,1,0),(0,0,-1)\}$$

2. Suppose you are starting at the equator and travelling due north, and your velocity is 5 m/sec. Compute the transport rates ρ_n and ρ_e associated with this velocity and heading. You may use the following constants: $\Omega_e = 7.2921151467 \times 10^{-5}$ rad/sec and $R_e = 6,378,137$ meters

Applications of Kalman Filtering to Simple Navigation Problems
Topics

- Static Distance Estimation

- Clock Democracy

Static Distance Estimation

- Scenario: Two soldiers each have a laser range finder. The first soldier points his range finder at a potential target and determines it to be 205.78 yards away. The second soldier determines with his range finder that the target is 201.22 yards away.

- How do these two soldiers combine their two measurements to produce an better estimate?

Static Distance Estimation

- Soldier A says, "Let's just take the average of our two measurements and call that in to the shooters."

- Soldier B disagrees and says, "No way! My laser range finder is much better than yours. We should just use my measurrement as the the best estimate."

Static Distance Estimation

- Actually, both soldiers are wrong!
- They should use a weighted average of both measurements as their final estimate.
- But what's the weighting scheme?

Static Distance Estimation

- Intuitively, the weighting scheme should take into account the relative uncertainty of each measurement.

- Also, if the two weights are equal, the weighted average should reduce to the simple unweighted average.

- Let r_a be the range measurment of Soldier A, r_b that of Solider B. Then, a weighted average would look like this:

$$\widehat{r} = w_a r_a + w_b r_b,$$

  where w_a is the weight associated with Solider A's measurement and w_b is that associated with Solider B's measurement.

Static Distance Estimation

- One way to set the weights using the uncertainty is to use the standard deviations associate with each measurement.

- Let σ_a be the standard deviation associated with laser range finder of Soldier A, σ_b that of Soldier B.

- Then we could set

$$w_a = \frac{\sigma_b^2}{\sigma_a^2 + \sigma_b^2} \quad \text{and} \quad w_b = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_b^2}.$$

Static Distance Estimation

- Why does $w_a = \dfrac{\sigma_b^2}{\sigma_a^2 + \sigma_b^2}$ have σ_b in its denominator instead of σ_a?

- Note also that these weights reduce to 1/2, as they should, when $\sigma_a = \sigma_b$.

Static Distance Estimation

- If we have several sequential measurmeents, from a variety of range measurement sources, we would want to put this into a recursive form like we did for the running average of the resistor.

- The recursive, weighted running average is beginning to look more and more like a Kalman filter.

Static Distance Estimation

- To see this, we have to first realize that the information or certainty (=inverse of uncertainty) of the weighted average estimate is:

$$1/p = 1/\sigma_a^2 + 1/\sigma_b^2.$$

- Note that p is the scalar version of the estimation error covariance matrix P and the $1/\sigma$ terms are the measurement error covariances R.

Static Distance Estimation

- This generalizes easily to k measurements (let's number our soldiers now from 1 to k):

$$1/p(k) = 1/\sigma_1 + \cdots + 1/\sigma_k = 1/p(k-1) + 1/\sigma_k$$

- The startling thing about this equation is that the infomration always increases, no matter how inaccurate the measurement! As long as the measurements are unbiased (i.e., zero mean) and Gaussian, it doesn't matter how inaccurate they are – each new measurement will convey some amount of additional information to improve our estimate.

Static Distance Estimation

- Finally, our recursive estimate of range $\widehat{r}(k)$ after the kth measurement is

$$\widehat{r}(k) = \widehat{r}(k-1) + K(k)(r(k) - \widehat{r}(k-1))$$

where

$$K(k) = p(k-1)\left[p(k-1) + \sigma_k^2\right]^{-1}$$

and

$$p(k) = \left[1/p(k-1) + 1/\sigma_k^2\right]^{-1}$$

Clock Democracy

- Coming soon ...

Day 3, Segment 2 (Continued)
Inertial Error Dynamics
Topics

- Imperfect Inertial Instruments
- From Strapdown Equations to Error Equations
- Initialization: Alignment

Imperfect Inertial Instruments

- Accelerometers and Gyros have inherent imperfections
- Bias, scale factor, g^2 terms, noise, random walks, etc.
- Inertial sensor clusters also have errors: misalignments
- Many error sources are calibrated out using rate tables during ATP
- Residual effects will influence strapdown nav equations

Imperfect Inertial Instruments

- Strapdown nav equations very sensitive to gyro errors
- Gyro biases yield erroneous C_B^L matrix which in turn causes erroneous accelerations to be computed in local level frame
- Thus, gyro errors affect not only attitude, but also position and velocity

Simple Inertial Error Models

- Gyros

$$\epsilon\omega_x = \Delta B_{gx} + \Delta K_{gx}\omega_x + \theta_{gxy}\omega_y + \theta_{gxz}\omega_z$$

$$\epsilon\omega_y = \Delta B_{gy} + \Delta K_{gy}\omega_y + \theta_{gyz}\omega_z + \theta_{gyx}\omega_x$$

$$\epsilon\omega_z = \Delta B_{gz} + \Delta K_{gz}\omega_z + \theta_{gzx}\omega_x + \theta_{gzy}\omega_y$$

where ΔB_{gi} is the ith gyro bias, ΔK_{gi} the ith gyro scale factor, and θ_{gij} is the misalignment of the ith gyro axis toward the jth gyro axis.

Simple Inertial Error Models

- Accelerometers

$$\epsilon a_x = \Delta B_{ax} + \Delta K_{ax} a_x + \theta_{axy} a_y + \theta_{axz} a_z$$

$$\epsilon a_y = \Delta B_{ay} + \Delta K_{ay} a_y + \theta_{ayz} a_z + \theta_{ayx} a_x$$

$$\epsilon a_z = \Delta B_{az} + \Delta K_{az} a_z + \theta_{azx} a_x + \theta_{azy} a_y$$

where ΔB_{ai} is the ith accelerometer bias, ΔK_{ai} the ith accelerometer scale factor, and θ_{aij} is the misalignment of the ith accelerometer axis toward the jth accelerometer axis.

Bias Models

- Bias in both types of instruments can be further broken down into several components:
  - fixed bias – constant from turn-on to turn-on – should be calibrated out
  - bias stability – varies from turn-on to turn-on – can't be calibrated out, but can be characterized in terms of variance – can model as random constant
  - bias drift – usually modeled as random walk

Scale Factor Models

- Scale factor can also be modeled as a sum of three components

- Often in nav filters, scale factor residual is modeled as a random constant

- Important to keep in mind that the contribution of scale factor error is a function of input (rate or acceleration)

Higher Order Models

- In expanding the output error as a Taylor series function of input, bias, scale factor, and misalignment show up as zero and first order terms.

- Higher order terms are also present, especially second order terms.

- In some calibration and alignment filters, such higher order effects are modelled

- SNU-84-1 includes error budgets associated with all the combined, higher order effects.

Taylor Series of Two Variables:

$$\begin{aligned} f(x,y) = f(x_0,y_0) &+ f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0) \\ &+ \frac{1}{2!}(f_{xx}(x_0,y_0)(x-x_0)^2 \\ &+ f_{xy}(x_0,y_0)(x-x_0)(y-y_0) \\ &+ f_{yy}(x_0,y_0)(y-y_0)^2) + \cdots \end{aligned}$$

Taylor Series provides basis for all error analysis.

From Strapdown Equations to Error Equations

- Nav equations require perfect inertial instruments
- Instruments are not perfect
- Inertial instrument error sources (bias, scale factor errors, etc.) generate erroneous rate and acceleration inputs to the nav equations
- Need to figure out the effects on the outputs of these equations

Strapdown Error Equations

- Nature of Errors
- Error Operator
- Error Equations

Strapdown Error Equations: Nature of Errors

- Errors propagate according to derived equations

- Errors are generally small, so approximating assumptions help simplify the math

  - linearize otherwise non-linear dynamics

  - small angle assumptions: $\sin\theta \approx \theta$

Strapdown Error Equations: Error Operator

If A is any quantity, we define error operator as follows:

$$\delta A = \widehat{A} - A, \tag{22}$$

where A is the true quantity of interest and $\widehat{A}$ is an apparent quantity. The latter could be a measured value, and estimated value, or some computed value. This means that

$$\widehat{A} = A + \delta A$$

i.e., the apparent value is the true value plus some error.

Strapdown Error Equations: Error Operator

Error operator satisfies:

(23) $$\delta(AB) = \delta A \cdot B + A \cdot \delta B$$

Proof:

$$\begin{aligned}
\delta(AB) &= \widehat{AB} - AB \\
&= \widehat{A}\widehat{B} - AB \\
&= (A + \delta A)(B + \delta B) - AB \\
&= AB + A \cdot \delta B + \delta A \cdot B + \delta A \cdot \delta B - AB \\
&= \delta A \cdot B + A \cdot \delta B
\end{aligned}$$

since we may assume that $\delta A \cdot \delta B \approx 0$.

Strapdown Error Equations

Applying these to the three differential equations, yields three error equations:

(24)

$$\delta\dot{C}_B^L = \delta C_B^L \Omega_{IB}^B + C_B^L \delta\Omega_{IB}^B - \left(\delta\Omega_{IE}^L + \delta\Omega_{EL}^L\right) C_B^L - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) \delta C_B^L$$

(25)

$$\delta\dot{v}^L = \delta C_B^L a_{\mathsf{sf}}^B + C_B^L \delta a_{\mathsf{sf}}^B + \delta g' - \left(2\delta\Omega_{IE}^L + \delta\Omega_{EL}^L\right) v^L - \left(2\Omega_{IE}^L + \Omega_{EL}^L\right) \delta v^L$$

(26)

$$\delta\dot{C}_L^E = \delta C_L^E \Omega_{EL}^L + C_L^E \delta\Omega_{EL}^L$$

Introducing Small Angular Rotations for Tilt and Position

- $\widehat{C}_B^L$, the computed (i.e., apparent) version of C_B^L is related additively to the true version by the error δC_B^L

$$\widehat{C}_B^L = C_B^L + \delta C_B^L$$

- It is also related to C_B^L by a small direction cosine matrix relating the local level frame to the computer frame $\widehat{L}$:

$$\widehat{C}_B^L = C_{\widehat{L}}^{\widehat{L}} C_B^L$$

Introducing Small Angular Rotations for Tilt and Position

- Substituting the second equation into the first allows us to solve for δC_B^L:

$$\begin{aligned}\delta C_B^L &= \widehat{C}_B^L - C_B^L \\ &= C_{\widehat{L}}^L C_B^L - C_B^L \\ &= (C_{\widehat{L}}^L - I) C_B^L\end{aligned}$$

  Now, let's introduce $\Phi = I - C_{\widehat{L}}^L$ to obtain:

$$\delta C_B^L = -\Phi C_B^L$$

Introducing Small Angular Rotations for Tilt and Position

- Using small angle approximations, $C_L^{\widehat{L}}$ can be written in the form

$$C_L^{\widehat{L}} = \begin{bmatrix} 1 & \phi_z & -\phi_y \\ -\phi_z & 1 & \phi_x \\ \phi_y & -\phi_x & 1 \end{bmatrix}.$$

  Thus, $\Phi = I - C_L^{\widehat{L}}$ is seen to be a skew-symmetric rotation matrix:

$$\Phi = \begin{bmatrix} 0 & -\phi_z & \phi_y \\ \phi_z & 0 & -\phi_x \\ -\phi_y & \phi_x & 0 \end{bmatrix}.$$

Introducing Small Angular Rotations for Tilt and Position

- Φ is therefore equal to the cross product operator associated with a vector $\phi^L = \begin{bmatrix} \phi_x^L & \phi_y^L & \psi_z^L \end{bmatrix}^\top$. I.e.,

$$\Phi = \left[\phi^L \times \right]$$

- The three components of ϕ^L will become three states in our inertial error dynamics Kalman filter.

- ϕ_x and ϕ_y are called tilts, while ψ_z is called the azimuth error.

Introducing Small Angular Rotations for Tilt and Position

- By a similar derivation, we can introduce a small angular position angle error $e^L = \begin{bmatrix} e_x^L & e_y^L & e_z^L \end{bmatrix}^\top$

- e^L is associated with δC_E^L just as ϕ^L is associated with δC_B^L:

$$\delta C_E^L = -E^L C_E^L,$$

  where E^L is the skew-symmetric matrix form of e^L.

- The three components of e^L yield the three position error states in a typical navigation Kalman filter.

Introducing Small Angular Rotations for Tilt and Position

- We are working to simplify Eq. (24):

$$\delta\dot{C}_B^L = \delta C_B^L \Omega_{IB}^B + C_B^L \delta\Omega_{IB}^B - \left(\delta\Omega_{IE}^L + \delta\Omega_{EL}^L\right) C_B^L - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) \delta C_B^L$$

- We have an expression for δC_B^L.

Introducing Small Angular Rotations for Tilt and Position

- Consider the earth rate error term $\delta\Omega^L_{IE}$. Since $\Omega^L_{IE} = C^L_E \Omega^E_{IE}$ and $\delta\Omega^E_{IE} = 0$, we derive:

$$\delta\Omega^L_{IE} = \delta C^L_E \Omega^E_{IE}.$$

- But $\delta C^L_E = -E^L C^L_E$. Therefore,

$$\begin{aligned} \delta\Omega^L_{IE} &= -E^L C^L_E \Omega^E_{IE} \\ &= -E^L \Omega^L_{IE}. \end{aligned}$$

Introducing Small Angular Rotations for Tilt and Position

- For the error in transport rate, $\delta\Omega_{EL}^{L}$, using the error operator on Eq. (14), a new error terms having to do with altitude error shows up:

$$\begin{aligned}\delta\omega_{EL}^{L} &= \delta\left[\frac{1}{R}u_r^L \times v^L\right] \\ &= \frac{1}{R}u_r^L \times \delta v^L - \frac{\delta h}{R^2}\left[u_r^L \times v^L\right]\end{aligned}$$

- Here, $R = R_e + h$ for a spherical earth of radius R_e.

Introducing Small Angular Rotations for Tilt and Position

- Eq. (24) can now be rewritten as:

  (27) $\dot{\Phi}^L = -\left(\Omega^L_{IE} + \Omega^L_{EL}\right)\Phi^L - C^L_B \delta\Omega^B_{IB} + \Omega^L_{IE} E^L + \delta\Omega^L_{EL}$

  where $\delta\Omega^L_{EL}$ is the skew-symmetric matrix form of

$$\delta\omega^L_{EL} = \frac{1}{R}\left(u^L_R \times \delta v^L\right) - \frac{\delta h}{R^2}\left(u^L_R \times v^L\right)$$

Introducing Small Angular Rotations for Tilt and Position

- Similarly, the velocity equation can be rewritten:

$$\delta\dot{v}^L = C_B^L \delta a_{\mathsf{sf}}^B + a_{\mathsf{sf}}^B \times \phi^L - \left(2\omega_{IE}^L + \omega_{EL}^L\right) \times \delta v^L + \\ - \left(\delta\omega_{EL}^L - 2e^L \times \omega_{IE}^L\right) \times v^L + \frac{2g\delta h}{R_e} u_R^L + \delta g^L$$

  while the equation for the position angle error e^L itself is

(28) $$\dot{e}^L = -\Omega_{EL}^L e^L + \delta\omega_{EL}^L$$

Further Simplifications Using Telescope Pointing Angle Error

- $\psi^L = \phi^L - e^L$

- Leads to:

(29) $$\delta\dot{R}^L = \delta v^L - \Omega^L_{EL}\delta R^L$$

and

$$\delta\dot{v}^L = C^L_B \delta a^B_{\mathsf{sf}} + a^B_{\mathsf{sf}} \times \psi^L - \frac{g}{R_e}\Delta R^L_H - \left(2\omega^L_{IE} + \omega^L_{EL}\right) \times \delta v^L +$$

(30) $$- \left(\delta\omega^L_{EL} - 2e^L \times \omega^L_{IE}\right) \times v^L + \frac{2g\delta h}{R_e} u^L_R + \delta g^L$$

Further Simplifications Using Telescope Pointing Angle Error

- Finally, the equations for the telescope pointing angle errors are:

$$\dot{\psi}^L = -\left(\Omega^L_{IE} + \Omega^L_{EL}\right)\psi^L - C^L_B \delta\omega^B_{IB} \tag{31}$$

Strapdown Error Equations

- Three error equations yield three vector quantities of interest
  - position error vector (3) δR derived from δC_L^E
  - velocity error vector (3) δV^L (or δv^L)
  - attitude error vector (3) ϕ_x, ϕ_y, ψ_z from δC_B^L

Strapdown Equations

- See Matlab / Simulink Models
  - Strapdown Navigation Equations: stravnav.mdl
  - Strapdown Navigation Error Equations: straperr.mdl

Initialization: Alignment

- Coarse Leveling
  - Gimbal system limited by platform torquing rates
  - Strapdown systems have no real rate limits
  - Time average of accel outputs scaled by $1/g$
  - Yields third row of DCM to begin fine alignment
- Fine Alignment
- Alignment filter aids in further leveling

Segment 11
The Basic 9-State Kalman Filter
Topics

- Whole State Filters vs. Error State Filters
- The Canonical Nine States
- INS Error State Dynamics and Transition Matrix

Whole State vs. Error State Filters

- Whole state filters ...
  - model whole position, velocity, attitude
  - deal with large and rapidly varying quantities
  - are highly non-linear

Whole State vs. Error State Filters

- Error state filters ...
  - remove some of the linearities by differencing outputs from aiding sensors with corresponding quantities from INS
  - exhibit slowly varying dynamics: easier to linearize
  - have smaller quantities that allow us to take advantage of small angle assmptions

The Canonical Nine States

- 3 position errors: $\delta R_x, \delta R_y, \delta R_z$ in local level coordinates
- 3 velocity errors: $\delta V_x, \delta V_y, \delta V_z$ in local level coordinates
- 3 attitude errors: ϕ_x, ϕ_y, ψ_z
  - ϕ_x, ϕ_y known as tilt errors
  - ψ_z known as azimuth error

Exercises

1. Rewrite equation (24)

$$\delta\dot{C}_B^L = \delta C_B^L \Omega_{IB}^B + C_B^L \delta\Omega_{IB}^B - \left(\delta\Omega_{IE}^L + \delta\Omega_{EL}^L\right) C_B^L - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) \delta C_B^L$$

to show the terms in the canonical order with position error, velocity error, and then attitude error.

2. Determine the 7th row of the system matrix F for the canonical 9-state error model. Use the cross-product operator matrix as required, e.g., $\Omega_{IE}^L = \left[\,\omega_{IE}^L\times\,\right] = \begin{bmatrix} 0 & \left[\omega_{IE}^L\right]_z & -\left[\omega_{IE}^L\right]_y \\ -\left[\omega_{IE}^L\right]_z & 0 & \left[\omega_{IE}^L\right]_x \\ \left[\omega_{IE}^L\right]_y & -\left[\omega_{IE}^L\right]_x & 0 \end{bmatrix}$.
Introduce simplifying notation! E.g., set $\alpha = \omega_{IE}^L$ Hint: Use telescope pointint angle error equation (31)

A 15-State Inertial Error Model: State Vector

$$x = \begin{bmatrix} \delta R_X \\ \delta R_Y \\ \delta V_X \\ \delta V_Y \\ \psi_X \\ \psi_Y \\ \psi_Z \\ \epsilon_x \\ \epsilon_y \\ \epsilon_z \\ \alpha_x \\ \alpha_y \\ \delta\theta_Z \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

A 15-State Inertial Error Model: State Dynamics Matrix

$$
\begin{bmatrix}
1 & 0 & T & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -V_Y T & 0 \\
0 & 1 & 0 & T & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -V_X T & 0 \\
\phi_{31} & \phi_{32} & 1 & 2\omega_Z T & 0 & -a_Z T & a_Y T & 0 & 0 & 0 & c_{11} T & c_{12} T & a_Y T & 0 \\
\phi_{41} & \phi_{42} & \phi_{43} & 1 & a_Z T & 0 & -a_X T & 0 & 0 & 0 & c_{21} T & c_{22} T & -a_X T & 0 \\
0 & 0 & 0 & 0 & 1 & \omega_Z T & -\omega_Y T & c_{11} T & c_{12} T & c_{13} T & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\omega_Z T & 1 & -\omega_X T & c_{21} T & c_{22} T & c_{23} T & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \omega_Y T & -\omega_X T & 1 & c_{31} T & c_{32} T & c_{33} T & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & e^{-T/\tau_g} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & e^{-T/\tau_g} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & e^{-T/\tau_g} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & e^{-T/\tau_a} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & e^{-T/\tau_a} & 0 & 0 \\
\frac{V_Y T}{R^2} & \frac{V_X T}{R^2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & e^{-T/} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
$$

where T is the delta time interval, τ_g,τ_a,τ_N,τ_E are Markov time corelation constants for gyro bias, accelerometer bias and GPS north and east biases, respectively.

Segment 12
Challenges of INS Filters
Topics

- Nine States Are Not Enough
- Not Constant Coefficient
- Non-Linearities: Extended Kalman Filter
- Observability

Nine States Are Not Enough

- The canoncial 9 states only characterize the inertial error dynamics
- They treat of the inner coupling between inertial error sources
- However, they don't help characterize dynamics of aiding sensor errors

Nine States Are Not Enough

- They don't help estimate aiding states such as Doppler bias and boresight errors
- Or GPS receiver clock errors
- They also don't help us to estimate inertial instrument drifts such as bias drift or scale factor errors

Not Constant Coefficient

- Notice that the F matrix that has emerged (implicitly) from the 9-state inertial error dynamics is not time invariant
- It is highly dependent on trajectory data such as acceleration and angular rate profiles
- These are functions of time
- Position dependent items such as latitude are also involved

Non-Linearities: Extended Kalman Filter

- When non-linearities are present, the F matrix emerges as Jacobian of non-linear system function f: $F(t) = \dfrac{\partial f}{\partial x}$

- If Jacobian is evaluated at a nominal value of the trajectory, we call this a linearized filter

- If Jacobian is evaluated at the previous optimal estimate, we call it an Extended Kalman filter

Observability

- Structural observability is concerned with how H maps measurements into states

- Will making measurements as governed by H actually enable us to estimate all of the states in the state vector?

Observability

- Observability means there is a linear path from the measurements to states

- Structural observability determined by Φ and H:

If H is $1 \times n$, where n is the size of the state vector, then the state x_0 is said to be **observable** from n scalar measurements $\{z_0, \ldots, z_{n-1}\}$ if the block matrix

$$\begin{bmatrix} H^\top & \Phi^\top H^\top & \cdots & (\Phi^\top)^{n-1} H^\top \end{bmatrix}$$

is of rank n

Observability

- Observability challenge for INS filters is usually a matter of choosing right navaids
- And also choosing the right measurement scheduling
- Also, certain kinds of divergence can be the result of poor observability (more on this later)

Exercises

- Use Eq. (29)

$$\delta \dot{R}^L = \delta v^L - \Omega^L_{EL} \delta R^L,$$

write the first three rows of the system dynamics matrix F in the continuous Kalman formulation:

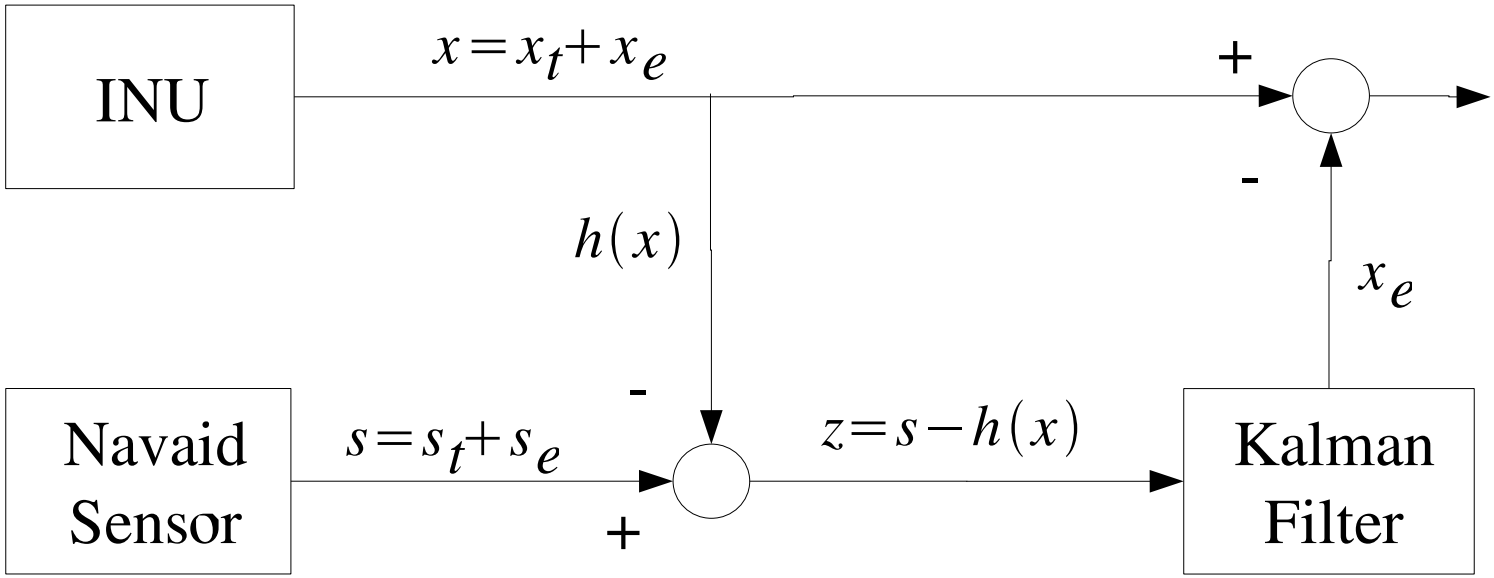
$$\dot{x} = Fx + w$$

Day 4, Segment 2
Survey of Navaids
Topics

- GPS
- Position Fixes
- Radar
- Doppler Velocity Sensor (DVS)
- External Tracking Aids

- Digital Terrain Elevation Data (DTED)

INS Error State Kalman Filter
Feedforward Mechanization



Note: $z = s - h(x) = s_t - h(x_t) + s_e - h(x_e)$

Navigation Aids (Navaids)

- Measurement / Observation sources that aid in estimating INS error sources and dynamics

- Because we are interested in estimating INS errors, we use differencing to cancel out the whole state information

- INS outputs $x = x_t + x_e$, where x_t is the true whole state information generated by INS, i.e., position, velocity, attitude, etc.

- Navaid sensor outputs $s = x_t + x_e$, where x_t is once again the true quantities corrupted by sensor errors x_e.

Navigation Aids (Navaids)

- Note that the navaid sensor usually only outputs part of the whole state vector, e.g., position and velocity, but not attitude (in the case of GPS)

- That's one reason why we have the h function

- The INS signal is mapped by a linearized measurement function (or matrix) h such that $h(x) = h(x_t) + h(x_e)$ and $h(x_t) = s_t$

Navigation Aids (Navaids)

- Thus, we have

$$\begin{aligned} z &= s - h(x) \\ &= s_t + s_e - h(x_t) - h(x_e) \\ &= s_t - h(x_t) + s_e - h(x_e) \\ &= h(x_t) - h(x_t) + s_e - h(x_e) \\ &= s_e - h(x_e) \end{aligned}$$

- Our measurement, therefore, is the difference between the navaid sensor errors and the INS errors

Navigation Aids (Navaids)

- If s_e is more than just white measurement noise, then we would need to model this vector in the filter according to known or assumed dynamics

- Hence, sometimes adding a new navaid means adding new states to your filter

Navigation Aids (Navaids)

- It is crucial to examine the innovations sequence for any non-white time correlations!

$$z_k - h\left[\widehat{x}^-(k)\right] = s_e(k) - h\left[(x_e(k)\right] - h\left[\widehat{x}^-(k)\right]$$

  where we have added the index k for clarity

- We should have $\mathsf{E}\left[(z_k - h\left[\widehat{x}^-(k)\right])(z_j - h\left[\widehat{x}^-(j)\right])^\top\right] = 0$ when $k \neq j$, which will in general be true provided $\mathsf{E}\left[s_e(k)\right] = 0$

- If this is not the case, then it is probably s_e that is the culprit and you'd better start modeling!

- The condition

$$\mathsf{E}\left[(z_k - h\left[\widehat{x}^-(k)\right])(z_j - h\left[\widehat{x}^-(j)\right])^\top\right] = 0$$

  says that the innovations sequence is uncorrelated in time.

- Recall that the innovations sequence is $\nu_k = z_k - h\left[\widehat{x}^-(k)\right]$

- Thus, we should have $\mathsf{E}\left[\nu_k \nu_j^\top\right] = 0$

- This means that the optimal estimator $\widehat{x}^+(k)$ always makes optimal use of the new information provided by ν_k

GPS: The Global Positioning System

- 24+ satellites orbiting at 10,900 miles above surface
- Three Segments: Control, Space, User
- Time-tagged signals and known satellite position enable calculation of signal time of flight and hence range traversed from satellite to receiver
- Range to four satellites required to determine time plus three position coordinates

GPS: The Global Positioning System

- Satellites have precise atomic clocks
- User set receiver has less precise clock
- Time is of the essence: comparing satellite-generated time-tag with user clock time of arrival determines transit time
- Note that user clock can be a source of systematic error

GPS: The Global Positioning System

- Three types of navaid coupling:
  - Loose coupling
  - Tight coupling
  - Deep coupling

Loosely-Coupled GPS

- GPS receiver-generated position and velocity are differenced with INS position and velocity
- GPS position and velocity are generally pre-filtered (with a Kalman filter) before being used in inertial filter
- GPS position and velocity are also composite quantities from all satellites
- Least accurate approach to using GPS as a navaid

Loosely-Coupled GPS

- The GPS receiver does not have the advantage of using inertial information to completely separate process noise from measurement noise

- Simplifies Kalman integration: Don't necessarily have to model GPS receiver clock errors (assume receiver has already done so)

Tightly-Coupled GPS

- Pseudorange and delta pseudorange for each individual satellite are direct measurements presented to the filter
- Contains only measurement noise, not receiver filter's model of process noise
- Gives INS filter more fine-grained control over GPS measurements
- Higher accuracy

Deeply-Coupled

- Uses high-rate inertial data to aid in code and carrier phase tracking loops

- Most accurate solution when combined with tightly-coupled INS filter as well

GPS Errors

- Ephemeris data: Satellite's reported position can be in error
- Satellite clock: Negligible since SA (selective availability) has been turned off
- Ionosphere: Pseudorange errors caused by free electrons in the ionosphere

GPS Errors (cont)

- Troposphere: Pseudorange errors casued by temperature, pressure, humidity influences on the speed of light

- Multipath: Errors casued by locking on to reflected signals

- Receiver Noise: Errors caused by receiver's thermal noise and software inaccuracies

Error Source	**C/A-Code Error Budget (meters)**	**P-Code Error Budget (meters)**
Multipath	12.0	1.2
Receiver Noise / Software	11.1	1.1
Ionosphereic Delays	9.0	3.1
Satellite Clock and Ephemeris Errors	3.9	3.9
Tropspheric Delays	2.0	2.0

Using Loosely-Coupled GPS Measurements

- Position
  - Form primary GPS position observation by subtracting Kalman-corrected INS reference position from raw GPS position
  - Next, the innovation (or measurement residual) is formed by subtracting off the extrapolated state estimate of position error
  - Extrapolated position error estimate needs to compensate for accumulated velocity error during Kalman cycle (or some fraction thereof)

An example (JOVIAL) from a transport aircraft navigation filter:

```
165  COMPUTE RESIDUALS
166      RESIDUALS(1) = Y'KAL'OBS(INU,1) * OBS'COUNT'SCALER - SVVAR
167  - 5.0 * SVVAR(4);
168      RESIDUALS(2) = Y'KAL'OBS(INU,2) * OBS'COUNT'SCALER - SVVAR
169  - 5.0 * SVVAR(5);
170      RESIDUALS(3) = Y'KAL'OBS(INU,7) * OBS'COUNT'SCALER - SVVAR
171          - 5.0 * SVVAR(6);
```

GPS Velocity Measurements

- Based on carrier phase measurements in receiver
- Generally independent of position measurements
- Observation is once again difference between Kalman-compensated INS reference velocity and GPS-indicated velocity

GPS Measurment Noise Covariance

- Based on EHE (estimated horizontal error) and EVE (estimated vertical error)

- These numbers highgly dependent on GDOPs

Geometric Dillution of Precision (GDOP)

- Satellites are not geostationary
- H matrix is time-varying
- Observability varies with time
- $GDOP = \sqrt{\text{tr}\left(H^\top H\right)^{-1}}$

GDOP Chimney

- Large DGOP is bad; normal values 3-5
- Singularity periods of time (about 1 hr) when GDOP values spike upward with a peak as much as 1000
- GDOP 50 $\approx$ 500 m position error
- Chimney's effects should be less acute if receiver has enough channels to track many satellites and to handle constellation changes

Position Fix

- Overfly Fix
- Known location
- Delta formed between overfly location and indicated, Kalman-compensated INS position
- Usually a one-shot opportunity, i.e., not periodic like GPS

Radar Fixes

- Instantaneous azimuth and elevation angles to a target of known latitude and longitude
- Or converted to local Cartesian coordinates (north, east, and altitude)
- Amount of cursor movement required to align curson with target yields position fix delta
- Otherwise, processed in filter like GPS or visual position fix

Doppler Velocity Sensor (DVS)

- DVS provides genuine velocity measurement with respect to ground

- DVS provides heading and drift velocities V_H and V_D

- Nonlinear measurement relationship to INS-indicated velocities V_x and V_y and platform azimuth β:

$$V_H = -V_x \sin\beta + V_y \cos\beta$$
$$V_D = V_x \cos\beta + V_y \sin\beta$$

DVS

- Relationship between (V_x, V_y) and (V_H, V_D) is non-linear h

- To generate a linearized H matrix we compute partial derivatives (Jacobians) to yield:

$$H = \begin{bmatrix} 0 & -\sin\beta & 0 & 0 & \cos\beta & 0 & 0 & 0 & -V_x\cos\beta - V_y\sin\beta \\ 0 & \cos\beta & 0 & 0 & \sin\beta & 0 & 0 & 0 & -V_x\sin\beta + V_y\cos\beta \end{bmatrix}$$

  for a canonical 9-state filter.

External Tracking Aids

- External platform tracks your vehicle and sends position / velocity info (JTIDS)

- Bearing and distance measurments
  - VOR, DME, TACAN

Digital Terrain Elevation Data (DTED)

- Published by National Geospatial Intelligence Agency (NGA)
  - formerly NIMA, formerly DMA
- 1 deg geocells of terrain height for each lat/long pair
- DTED Levels 0,1,2

DTED

- As navaid, a trail of radar altimeter measurements are stored
- Trail is correlated with DTED to determine offset
- SITAN, TERPROM